

Yifan Zhang

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South China University of Technology, Guangzhou, China

RESEARCH PROFILE

Ph.D. student in Software Engineering studying memory enhancement and reliability evaluation for large language model-based agents in scientific research. My work links three areas: multi-source memory enhancement for scientific tasks, reliability evaluation of memory-driven behavior, and construction and validation of domain agents in representative scientific tasks.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=SUBMITTED, A=PREPRINT, T=THESIS, *=EQUAL CONTRIBUTION

- [S.1] Yifan Zhang, Ziyang Li, Xiaojing Zhang, Qian Tao (2026). **MemCasa: Can LLM Agents Act on What They Remember?** Manuscript under review.
- [S.2] Yifan Zhang, Qian Tao (2026). **Beyond Field Accuracy: Two-Axis Diagnosis of Inverse-PINN Parameter Error.** Manuscript under review.
- [J.1] Sabine Brunswicker, Yifan Zhang, Christopher Rashidian, Daniel W. Linna Jr. (2025). **Trust through Words: The Systemize-Empathize-Effect of Language in Task-oriented Conversational Agents.** *Computers in Human Behavior*, 165, 108516.
- [A.1] Shiwen Ni, Guhong Chen, Shuaimin Li, Xuanang Chen, Siyi Li, Bingli Wang, Qiyao Wang, Xingjian Wang, Yifan Zhang, Liyang Fan, Chengming Li, Ruifeng Xu, Le Sun, Min Yang (2025). **A Survey on Large Language Model Benchmarks.** *arXiv:2508.15361*.
- [C.1] Tianlu Zheng*, Yifan Zhang*, Xiang An, Ziyong Feng, Kaicheng Yang, Qichuan Ding (2025). **Gradient-Attention Guided Dual-Masking Synergetic Framework for Robust Text-based Person Retrieval.** *EMNLP 2025*.
- [J.2] Yifan Zhang, Christopher Radishian, Sabine Brunswicker, Dan Whitenack, Daniel W. Linna Jr. (2024). **Empathetic Language in LLMs under Prompt Engineering: A Comparative Study in the Legal Field.** *Procedia Computer Science*, 244, 308–317.
- [C.2] Sabine Brunswicker, Yifan Zhang, Christopher Rashidian, Daniel W. Linna Jr. (2024). **The Impact of Empathy Display in Language of Conversational AI: A Controlled Experiment with a Legal Chatbot.** *HICSS-57*, 455–466.
- [A.2] Xin Wang*, Yifan Zhang*, Xiaojing Zhang, Longhui Yu, Xinna Lin, Jindong Jiang, Bin Ma, Kaicheng Yu (2024). **PatentAgent: Intelligent Agent for Automated Pharmaceutical Patent Analysis.** *arXiv:2410.21312*.
- [A.3] Xin Wang, Xiaojing Zhang, Yifan Zhang, Xinna Lin, Ruichen Li, Kaicheng Yu (2024). **An Empirical Study on Reconstructing Scientific History to Forecast Future Trends.** OpenReview, 2024.
- [T.1] Yifan Zhang (2024). **Advancements in Chest Radiography Pneumonia Classification through Fine-Tuning using MIMIC-CXR-JPG Dataset.** Master's thesis, Northwestern University.

EDUCATION

- **South China University of Technology** Sept 2025 - June 2029 (Expected)
Doctor of Philosophy in Software Engineering Guangzhou, China
 - Advisor: Prof. Qian Tao
 - Dissertation research: memory enhancement and reliability evaluation for LLM-based agents in scientific research
- **Northwestern University** Sept 2022 - April 2024
Master of Science in Computer Science Evanston, US
- **Colby College** May 2022
Bachelor of Arts in Computer Science, Minor in Managerial Economics Waterville, US
 - Presidential Scholarship recipient

EXPERIENCE

- South China University of Technology [🌐]

PhD Student

◦ Conducting doctoral research under Prof. Qian Tao on an evidence–memory–behavior–scientific–outcome pipeline for reliable LLM-based scientific agents.

◦ Led two first-author manuscripts on memory-in-action evaluation for LLM agents and diagnostic reliability evaluation for inverse physics-informed neural networks.

Sept 2025 - Present

Guangzhou, China
- DeepGlint [🌐]

MLE Intern

◦ Co-developed **HMDocRAG-Bench**, a hierarchical benchmark with 6,000 question–answer pairs for evaluating document retrieval-augmented generation systems.

◦ Engineered multi-hop, multimodal, and multi-page data-generation pipelines and contributed to the five-million-sample **WebPerson** dataset used in GA-DMS (EMNLP 2025).

March 2025 - Sept 2025

Beijing, China
- AutoLab at Westlake University [🌐]

Visiting Student

◦ Co-led **PatentAgent** for automated pharmaceutical patent analysis and contributed to agent-based scientific knowledge synthesis and trend forecasting.

June 2024 - March 2025

Hangzhou, China

SELECTED RESEARCH PROJECTS

Agent Memory and Scientific Reliability

- MemCasa

LLM Agents, Memory Systems, Benchmarking

◦ Led the design and evaluation of a deterministic suite of 840 evaluation units that tests memory through executable behavior across six memory demands, with strict and graded scoring.

◦ Compared controlled memory policies under matched model and simulator settings, using deterministic validators to distinguish exact success from partial competence.

2026 - Present

(Manuscript under review)
- Beyond Field Accuracy / Inverse-PINN Reliability

Physics-Informed Neural Networks, Scientific Reliability, Diagnostics

◦ Introduced a two-axis diagnosis separating finite-sample observation resolution from learned-field residual preference; under each run’s frozen final adaptive-weight residual view, diagnosed displacement tracked returned signed log-parameter error across 240 fresh-noise retrainings ($r = .994$).

2025 - Present

(Manuscript under review)
- Scientific-Memory Reliability Pilot

LLM Agents, Scientific Workflows, Reliability Evaluation

◦ Completed a single-model exploratory 24-case, 288-decision inverse-PINN memory study whose benefit gate was prespecified and frozen before the full run but was not met; qualitative audit exposed a gap between recognizing provenance conflict and executing the required safe action.

2026

(Completed frozen pilot)

Domain Agents for Scientific Tasks

- PatentAgent

LLM Agents, Retrieval-Augmented Generation, Tool Use

◦ Co-developed an agent workflow for automated pharmaceutical patent analysis, including evidence retrieval, tool-assisted analysis, and structured reporting.

2024

(arXiv:2410.21312)

TEACHING EXPERIENCE

- Project Manager, Department of Computer Science

CS396: Intro to Game Design

Fall 2023

Northwestern University
- Teaching Assistant, Department of Mathematics

MA 125: Single-Variable Calculus

Spring 2020

Colby College
- Teaching Assistant, Department of Computer Science

CS 15X: Computational Thinking

Fall 2019

Colby College

SKILLS

- **Programming Languages:** Python, Julia, R, MATLAB, C/C++, Java, SQL
- **Data Science & Machine Learning:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy
- **Specialized Areas:** LLM Agents, Agent Memory, Reliability Evaluation, AI for Science, NLP
- **Research Skills:** Experimental Design, Benchmark Construction, Intervention-Based and Diagnostic Evaluation, Scientific Machine Learning, Technical Writing

ADDITIONAL INFORMATION

Languages: Mandarin (Native), English (Fluent), Spanish (Intermediate)
Interests: Run (Half-Marathon), Music (Flute in Jazz Band)